Krishna Mula

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Education

San Jose State University — San Jose, CA
Master of Science in Computer Science — *Aug 2025 - May 2027*
Coursework: Distributed Systems, Advanced Parallel Processing, Cloud Computing, Database Systems, Machine Learning.

Experience

Intake AI — San Francisco, CA
Software Engineer Intern — *Jun 2026 - Present*

- Introduced **cross-encoder reranking** to the chat assistant clinical trial sites use to query their trial documents, raising **recall@5 from 64% to 78%**, and shipped it as a GPU service (FastAPI, fp16, NVIDIA L4).
- Cut **reranking p50 latency 33%** (986 to 662 ms) and **peak GPU memory 46%** by reducing the inference batch size from 32 to 4, taking over 300 ms off every chat response.
- Shrank the **document-processing service's image from 11.8 GB to 908 MB**, cutting image pull time for each new task from **105 s to 7 s**.
- Split production from a single EC2 host into **independent ECS services** running multiple replicas each, with **event-driven workers autoscaling on SQS queue depth**, removing the single point of failure.

PwC
Software Engineer — *Aug 2022 - Jul 2025*

- Architected a **RAG pipeline** for IT security compliance, embedding documents into ChromaDB for semantic retrieval and integrating **LLM inference** for automated validation against regulatory standards, **reducing review time by 70%**.
- Designed a **RabbitMQ-backed async ingestion architecture** that decoupled upload APIs from parsing, chunking, embedding, and vector indexing, **reducing latency by 30%** under bursty workloads.
- Built an ETL pipeline using PyRFC to extract SM20/SM21 security audit logs from **5 production SAP systems**, replacing a process that required **2+ hours of manual export and formatting** per report.
- Automated security controls execution on client SAP systems by codifying **25+ controls** as Flask + MongoDB REST APIs, ingesting data directly from SAP HANA and validating against security specifications.
- Built **serverless compliance reporting APIs on AWS Lambda** for 15+ client systems, pre-warmed to handle twice-daily traffic bursts without cold-start delays.

Software Engineering Intern — *Feb 2022 - Aug 2022*

- Built Nivo.js dashboards tracking security control pass/fail rates, compliance coverage, and historical trends, letting teams prioritize which controls to automate next.
- Automated deployments with **Azure CI/CD** pipelines, reducing release cycles from **2 days to 4.5 hours**.

Projects

Aether | *Python · TCP Sockets · FastAPI · Docker SDK · OpenTelemetry · Prometheus · D3.js*

- Built a **distributed pub-sub messaging system** over raw TCP with a gossip-based multi-broker mesh, UUID deduplication, configurable fanout/TTL, and heartbeat-driven peer eviction.
- Implemented **Chandy-Lamport global snapshots** with replication to k peers driving hybrid broker failover; a fresh snapshot triggers full state recovery and a stale one redistributes subscribers to least-loaded brokers.
- Built a **FastAPI orchestration control plane** over the Docker SDK with a pull-based subscriber assignment registry as the source of truth during recovery; subscribers detect failure via Ping/Pong and self-heal without coordinated restarts.

Technical Skills

Languages: Python, TypeScript, SQL, C++.
ML & Inference: PyTorch, Hugging Face Transformers, cross-encoder reranking, fp16 GPU serving, ChromaDB.
Frameworks: FastAPI, Flask, Node.js, React, Next.js.
Infrastructure: Terraform, Docker, AWS (ECS, Fargate, EC2, ECR, Lambda, SQS, S3, RDS, DynamoDB, CloudWatch, IAM/OIDC), Azure (Functions, VM, Entra ID), GitHub Actions, Nginx.
Data & Messaging: PostgreSQL, MongoDB, Redis, Cassandra, SQLite, RabbitMQ, SQS.
Observability: OpenTelemetry, Prometheus, Grafana, Loki, Sentry.